

CSCI5527 Homework 0

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1 Problem 1

1.1 Part a

$$(A^T B)_{ij} = \sum_{k=1}^m A_{ki} B_{kj}$$
$$\text{tr}(A^T B) = \sum_{i=1}^n (A^T B)_{ii} = \sum_{i=1}^n \sum_{j=1}^m A_{ji} B_{ji}$$

$$\langle A, B \rangle = \text{tr}(A^T B)$$

For the Frobenius norm:

$$\|M\|_F = \sqrt{\langle M, M \rangle}$$

$$\langle M, M \rangle = \text{tr}(M^T M)$$

$$\|M\|_F = \sqrt{\text{tr}(M^T M)}$$

1.2 Part b

$$\langle A, B \rangle = \text{tr}(A^T B)$$

The inner product is symmetric:

$$\langle A, B \rangle = \langle B, A \rangle.$$

By definition:

$$\langle B, A \rangle = \text{tr}(B^T A).$$

So,

$$\text{tr}(A^T B) = \text{tr}(B^T A).$$

1.3 Part c

$$\operatorname{tr}(AB^T) = \sum_i (AB^T)_{ii}.$$

$$(AB^T)_{ii} = \sum_j A_{ij}B_{ij}.$$

$$\operatorname{tr}(AB^T) = \sum_i \sum_j A_{ij}B_{ij}.$$

$$\langle A, B \rangle = \sum_i \sum_j A_{ij}B_{ij},$$

$$\operatorname{tr}(AB^T) = \langle A, B \rangle.$$

$$\langle B, A \rangle = \operatorname{tr}(B^T A).$$

Since we established $\operatorname{tr}(AB^T) = \langle A, B \rangle$ and we know $\langle A, B \rangle = \langle B, A \rangle$, we substitute:

$$\operatorname{tr}(AB^T) = \operatorname{tr}(B^T A).$$

1.4 Part d

From part c,

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Define $X = M_1 M_2$ and $Y = M_3$, so that

$$\operatorname{tr}(M_1 M_2 M_3) = \operatorname{tr}(XY).$$

Applying $\operatorname{tr}(XY) = \operatorname{tr}(YX)$,

$$\operatorname{tr}(M_1 M_2 M_3) = \operatorname{tr}(M_3 M_1 M_2).$$

Next, set $X = M_3 M_1$ and $Y = M_2$, so that

$$\operatorname{tr}(M_3 M_1 M_2) = \operatorname{tr}(M_2 M_3 M_1).$$

Thus,

$$\operatorname{tr}(M_1 M_2 M_3) = \operatorname{tr}(M_3 M_1 M_2) = \operatorname{tr}(M_2 M_3 M_1).$$

1.5 Part e

From part a,

$$\langle ACB, D \rangle = \text{tr}((ACB)^T D).$$

$$\langle ACB, D \rangle = \text{tr}(B^T C^T A^T D).$$

By the cyclic property of the trace,

$$\text{tr}(B^T C^T A^T D) = \text{tr}(C^T A^T D B^T).$$

$$\text{tr}(C^T A^T D B^T) = \langle CB, A^T D \rangle.$$

$$\langle ACB, D \rangle = \langle CB, A^T D \rangle.$$

Applying the cyclic property again,

$$\text{tr}(B^T C^T A^T D) = \text{tr}(A^T D B^T C^T).$$

$$\text{tr}(A^T D B^T C^T) = \langle AC, D B^T \rangle.$$

$$\langle ACB, D \rangle = \langle CB, A^T D \rangle = \langle AC, D B^T \rangle.$$

1.6 Part f

Using the Frobenius norm definition,

$$\|M\|_F = \sqrt{\text{tr}(M^T M)}.$$

$$M^T M = (U \Sigma V^T)^T (U \Sigma V^T).$$

$$M^T M = V \Sigma^T U^T U \Sigma V^T.$$

Since $U^T U = I$,

$$M^T M = V \Sigma^T \Sigma V^T.$$

By the cyclic property of the trace,

$$\text{tr}(M^T M) = \text{tr}(V \Sigma^T \Sigma V^T).$$

$$\text{tr}(V \Sigma^T \Sigma V^T) = \text{tr}(\Sigma^T \Sigma V^T V).$$

Since $V^T V = I$,

$$\text{tr}(\Sigma^T \Sigma V^T V) = \text{tr}(\Sigma^T \Sigma).$$

Since Σ is diagonal with singular values σ_i , we have

$$\Sigma^T \Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_r^2).$$

$$\text{tr}(\Sigma^T \Sigma) = \sum_{i=1}^r \sigma_i^2.$$

$$\|M\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2}.$$

1.7 Part g

Using the definitions of the inner product and the Hadamard product,

$$\langle A \odot B, C \rangle = \sum_{i,j} (A \odot B)_{ij} C_{ij} = \sum_{i,j} A_{ij} B_{ij} C_{ij}.$$

$$\sum_{i,j} A_{ij} (B \odot C)_{ij} = \langle A, B \odot C \rangle.$$

$$\sum_{i,j} B_{ij} (A \odot C)_{ij} = \langle B, A \odot C \rangle.$$

$$\langle A \odot B, C \rangle = \langle A, B \odot C \rangle = \langle B, A \odot C \rangle.$$

2 Problem 2

Here is a link to the confirmations for my derivations in parts a, b and c using ChatGPT: <https://chatgpt.com/share/67a98a77-96f4-8012-9984-135550155a4f>

2.1 Part a

$$f(B + \delta B) = \sum_{i=1}^N \left[y_i \langle B + \delta B, X_i \rangle - \log \left(1 + e^{\langle B + \delta B, X_i \rangle} \right) \right].$$

$$f(B + \delta B) = \sum_{i=1}^N \left[y_i \langle B, X_i \rangle + y_i \langle \delta B, X_i \rangle - \log(1 + e^{\langle B, X_i \rangle + \langle \delta B, X_i \rangle}) \right].$$

Using the given Taylor expansion,

$$\log(1 + e^{\langle B, X_i \rangle + \langle \delta B, X_i \rangle}) = \log(1 + e^{\langle B, X_i \rangle}) + \frac{e^{\langle B, X_i \rangle}}{1 + e^{\langle B, X_i \rangle}} \langle \delta B, X_i \rangle + O(\|\delta B\|^2).$$

$$f(B + \delta B) - f(B) = \sum_{i=1}^N \left(y_i - \frac{e^{\langle B, X_i \rangle}}{1 + e^{\langle B, X_i \rangle}} \right) \langle \delta B, X_i \rangle + O(\|\delta B\|^2).$$

So,

$$\nabla f(B) = \sum_{i=1}^N (y_i - p(X_i; B)) X_i,$$

2.2 Part b

$$f(B + \delta V) = \sum_{i=1}^N \left[y_i \langle B + \delta V, X_i \rangle - \log \left(1 + e^{\langle B + \delta V, X_i \rangle} \right) \right].$$

$$f(B + \delta V) = \sum_{i=1}^N \left[y_i \langle B, X_i \rangle + \delta y_i \langle V, X_i \rangle - \log \left(1 + e^{\langle B, X_i \rangle + \delta \langle V, X_i \rangle} \right) \right].$$

Using the given Taylor expansion,

$$\log(1 + e^{z+\delta}) = \log(1 + e^z) + \frac{e^z}{1 + e^z} \delta + \frac{1}{2} \frac{e^z}{(1 + e^z)^2} \delta^2 + O(\delta^3),$$

with $z = \langle B, X_i \rangle$ and $\delta = \langle V, X_i \rangle$,

$$\log(1 + e^{\langle B, X_i \rangle + \delta \langle V, X_i \rangle}) = \log(1 + e^{\langle B, X_i \rangle}) + \frac{e^{\langle B, X_i \rangle}}{1 + e^{\langle B, X_i \rangle}} \delta \langle V, X_i \rangle + \frac{1}{2} \frac{e^{\langle B, X_i \rangle}}{(1 + e^{\langle B, X_i \rangle})^2} \delta^2 \langle V, X_i \rangle^2 + O(\delta^3).$$

$$f(B + \delta V) - f(B) = \sum_{i=1}^N \left[y_i \delta \langle V, X_i \rangle - \frac{e^{\langle B, X_i \rangle}}{1 + e^{\langle B, X_i \rangle}} \delta \langle V, X_i \rangle - \frac{1}{2} \frac{e^{\langle B, X_i \rangle}}{(1 + e^{\langle B, X_i \rangle})^2} \delta^2 \langle V, X_i \rangle^2 + O(\delta^3) \right].$$

$$f(B + \delta V) - f(B) = \delta \sum_{i=1}^N (y_i - p(X_i; B)) \langle V, X_i \rangle - \frac{\delta^2}{2} \sum_{i=1}^N p(X_i; B) (1 - p(X_i; B)) \langle V, X_i \rangle^2 + O(\delta^3).$$

Comparing with the second-order Taylor expansion,

$$\langle V, \nabla^2 f(B) [V] \rangle = \sum_{i=1}^N p(X_i; B) (1 - p(X_i; B)) \langle V, X_i \rangle^2.$$

So,

$$\nabla^2 f(B) [V] = \sum_{i=1}^N p(X_i; B) (1 - p(X_i; B)) X_i X_i^T V.$$

2.3 Part c

For a perturbation $W \rightarrow W + \Delta$,

$$\sigma((W + \Delta)x) = \sigma(Wx) + \sigma'(Wx) \odot (\Delta x) + o(\|\Delta\|_F).$$

$$g(W + \Delta) = \|y - \sigma(Wx) - \sigma'(Wx) \odot (\Delta x)\|_2^2.$$

Using the identity,

$$\|a + b\|_2^2 = \|a\|_2^2 + \|b\|_2^2 + 2\langle a, b \rangle,$$

with $a = y - \sigma(Wx)$, $b = -\sigma'(Wx) \odot (\Delta x)$, we get

$$g(W + \Delta) = \|y - \sigma(Wx)\|_2^2 + \|\sigma'(Wx) \odot (\Delta x)\|_2^2 - 2\langle y - \sigma(Wx), \sigma'(Wx) \odot (\Delta x) \rangle + o(\|\Delta\|_F).$$

Ignoring higher-order terms,

$$g(W + \Delta) - g(W) = -2\langle y - \sigma(Wx), \sigma'(Wx) \odot (\Delta x) \rangle + o(\|\Delta\|_F).$$

$$-2 \sum_i ([y - \sigma(Wx)]_i [\sigma'(Wx) \odot (\Delta x)]_i).$$

$$-2\langle y - \sigma(Wx), \sigma'(Wx) \odot (\Delta x) \rangle = -2\langle \sigma'(Wx) \odot (y - \sigma(Wx)), \Delta x \rangle.$$

Since this holds for any small Δ , we conclude

$$\nabla g(W) = -2(\sigma'(Wx) \odot (y - \sigma(Wx)))x^T.$$

For $\sigma(x) = \sin(x)$, its derivative is $\sigma'(x) = \cos(x)$, so

$$\nabla g(W) = -2(\cos(Wx) \odot (y - \sin(Wx)))x^T.$$

2.4 Part d

$$\|y - Ax\|_2^2 = (y - Ax)^T (y - Ax).$$

$$\nabla f(x) = \nabla \left(\frac{1}{2}(y - Ax)^T (y - Ax) + \lambda x^T x \right).$$

Using $\nabla_x \frac{1}{2}\|Ax - y\|_2^2 = A^T(Ax - y)$ and $\nabla_x \lambda\|x\|_2^2 = 2\lambda x$, we get:

$$\nabla f(x) = A^T(Ax - y) + 2\lambda x.$$

To solve for x , set,

$$A^T(Ax - y) + 2\lambda x = 0.$$

$$A^T Ax + 2\lambda x = A^T y.$$

$$(A^T A + 2\lambda I)x = A^T y.$$

The matrix $A^T A + 2\lambda I$ is symmetric and positive definite. Since $A^T A$ is positive semi-definite and $2\lambda I$ is strictly positive definite, their sum remains positive definite:

$$v^T(A^T A + 2\lambda I)v > 0, \quad \forall v \neq 0.$$

Since all positive definite matrices are invertible, the system has a unique solution:

$$x^* = (A^T A + 2\lambda I)^{-1} A^T y.$$

2.5 Part e

If all eigenvalues of $A^T A + 2\lambda I$ remain positive, then the matrix remains invertible, and a unique minimizer still exists.

If some eigenvalues become zero, then the system has infinitely many solutions.

If some eigenvalues become negative, then $f(x)$ is no longer convex and does not necessarily have a minimizer.

3 Problem 3

3.1 Part a

Yes, this can be calculated using Bayes' theorem.

$$P(\text{drinker} \mid \text{cancer}) = P(\text{cancer} \mid \text{drinker})P(\text{drinker}) / P(\text{cancer}) = 0.4 \cdot 10 / (0.4 \cdot 10 + 0.6) = 4 / 4.6 = 0.8696$$

3.2 Part b

$$P(\text{disease} \mid \text{positive}) = P(\text{positive} \mid \text{disease})P(\text{disease}) / P(\text{positive}) = .001 * .99 / (.001 * .99 + 0.999 * 0.2) = 0.0472$$